Lab 8: Hubble's Law

Apr 1, 2024

1 Introduction: The Expanding Universe

In 1929, Edwin Hubble made a discovery that would change the course of astronomy forever. While poring over data on the distances and velocities of faraway galaxies, Hubble noticed two things: one, that almost all galaxies are systematically moving *away* from us; and two, that galaxies that are *further* away from us are moving away *faster*. From these observations, Hubble drew a simple – but groundbreaking – conclusion: **space itself is expanding**.

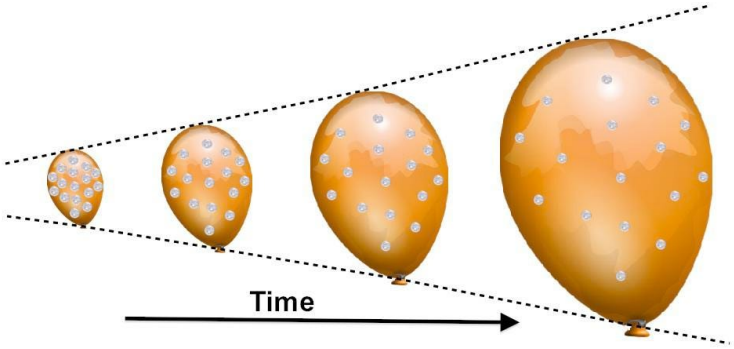


Figure 1: An expanding balloon analogizing the expanding Universe.

Consider Figure 1, showing a balloon covered in stickers. While these stickers themselves aren't moving, the expansion of the balloon pushes them apart. Now, imagine that these stickers are galaxies and that the balloon is the expanding Universe. While a galaxy may not have its own intrinsic velocity, it'll still appear to move away from us due to the expansion of space itself. Therefore, distant galaxies acquire a *redshift* due solely to cosmological expansion.

If the Universe is *expanding*, then it must have a finite age; at some time in the distant past, the Universe was just a point. Hubble's discovery thus revolutionized the field of cosmology – the study of the history and composition of the Universe – paving the way for theories of the Big Bang, of galaxy formation and evolution, of the birth of the first stars, and so much more.

In today's lab, we will trace how our understanding of Hubble's discovery has evolved over time. We will also learn how to measure the expansion rate of the Universe via the *cosmic distance ladder*. This lab should leave you with a deeper understanding of *Hubble's law*, which forms the foundation of modern cosmology.

2 Hubble Trouble: Finding the Expansion Rate of the Universe

Mathematically, the expansion of the Universe is encoded in **Hubble's Law**,

$$v = H_0 D, \tag{1}$$

which states that the velocity, v, of an object moving due to the expansion of space is related to the object's distance, D, via some constant, H_0. H_0, or the **Hubble constant**, therefore tells us *how fast* the Universe is expanding. According to equation 1, we can find this expansion rate by making a plot of velocity vs. distance (for many distant astronomical objects) and measuring the *slope* of the line that best fits the data. In this section, we'll see how this works out in practice.

2.1 Retracing Hubble's steps

Let's travel back in time to 1929, when Hubble was working out his theory of the expanding Universe. Given that, just a few years prior, Hubble had shown that other galaxies were in fact *outside of* the Milky Way, we can assume that telescope data in the 1920s wasn't fantastic.

Follow this link to get the data... copy the data into your own sheet, please don't make changes to this sheet!: `https://docs.google.com/spreadsheets/d/1XoU23LoK2KbG7E3LYdwePd8Byy_22TlKafQFMxP5yDg/edit?usp=sharing` the first tab (labeled "1929") shows Hubble's original data for 24 galaxies. Note that the velocity is labeled "recession velocity" – this means that we're looking only at motions towards or away from us. **Be sure to include the appropriate graphs in your lab write-up.**

1. Make a scatter plot of Hubble's data, putting recession velocity (in km/s) on the vertical axis and distance (in Mpc) on the horizontal axis. For reference, 1 Mpc ("megaparsec") is about 3 million light years, or about 3×10^{19} kilometers.

2. On the same figure, plot the line that best fits the data. In Google Sheets, you can do this by going to the "Customize" tab under the chart editor, clicking "Series," and then checking the box labeled "Trendline."

   (a) Hubble's law predicts a linear trend between recession velocity and distance. How well does your trend line fit the data? Are you convinced that there's actually a linear trend in the data?

   (b) Why doesn't the line perfectly fit the data? That is, what are some sources of error or uncertainty that could be causing deviation from the predicted linear model?

   (c) Does your line pass through the origin (i.e., does your line pass through the point of zero velocity and zero distance)? Do you expect the line to pass through the origin? (Hint: think about Equation 1)

   (d) If your line doesn't pass through the origin, *why* doesn't it?

3. Slightly below the check box for "Trendline," there's a drop-down menu for "Label;" set this to "Use Equation," and enable the check-box labeled "Show R^2" below it.

(a) For this best-fit line, what should be the *units* of the slope? (Hint: remember that slope is "rise over run," or the vertical change divided by the horizontal change)

(b) The equation of the best-fit line should be of the form $m \times x + b$, where m is the slope and b is the y-intercept. What is the slope of your best-fit line (remember to add units)? Congrats – you've just measured the Hubble constant!

(c) What value do you get for R^2? R^2 is a measure of how good the line fits the data, with $R^2 = 1$ being a perfect fit. Would you characterize this linear model as a "good" fit?

4. Since the Hubble constant tells us how fast the Universe is (and has been) expanding, it also tells us how *old* the Universe is. We can obtain a rough estimate for the age of the Universe by taking the reciprocal of the Hubble constant, or $1/H_0$.

   (a) The Hubble constant has weird units, so we'll first have to do some unit conversion. $1\,\frac{\text{km/s}}{\text{Mpc}} = 1.02 \times 10^{-12} \frac{1}{\text{year}}$. Use this to convert your measurement of H_0 to units of $\frac{1}{\text{year}}$.

   (b) Now, compute $1/H_0$ to obtain the age of the Universe in years. Does your answer make sense?

   (c) Using radiometric dating of rocks, geologists have determined the age of the Earth to be about 4.5×10^9 years old. How does this compare to your estimate of the age of the Universe? Does this make sense?

5. The objects from Hubble's experiment are all local! 2 Mpc is very nearby and we expect objects this close to be gravitationally bound (one consequence of this is that gravity is opposing the expansion of the universe). How do you think this effects our estimate on the age of the universe? And what further data would you need to obtain a better estimate?

2.2 The limits of Hubble's law

In the late 1990s, there was fierce competition between the Supernova Cosmology Project (lead by Saul Perlmutter at Berkeley) and the High-Z Supernova Search Team (led by Brian Schmidt at Harvard and Adam Riess at the Space Telescope Science Institute) to measure the most distant Type Ia supernovae possible, and thus to produce the highest-quality measurements of the Hubble constant. Both teams stumbled upon some strange results.

Figure 2 shows the Supernova Cosmology Project's data alongside older supernova data taken at shorter distances; here, magnitude is a proxy for distance and redshift is a proxy for velocity, so the axes are actually *flipped* compared to what we've been plotting.

6. Describe the trend in Figure 2 at large distances (i.e., at large magnitude and redshift). Does the data still follow the linear trend predicted by Hubble's law?

7. According to Hubble's law, objects that are more distant from us are receding from us faster, and thus the light emitted by more distant objects will be more redshifted. Light from more distant objects also takes longer to reach us, so more distant objects appear to be younger, or to have existed earlier in the Universe. In this sense, how can we use redshift as a proxy for cosmic time?

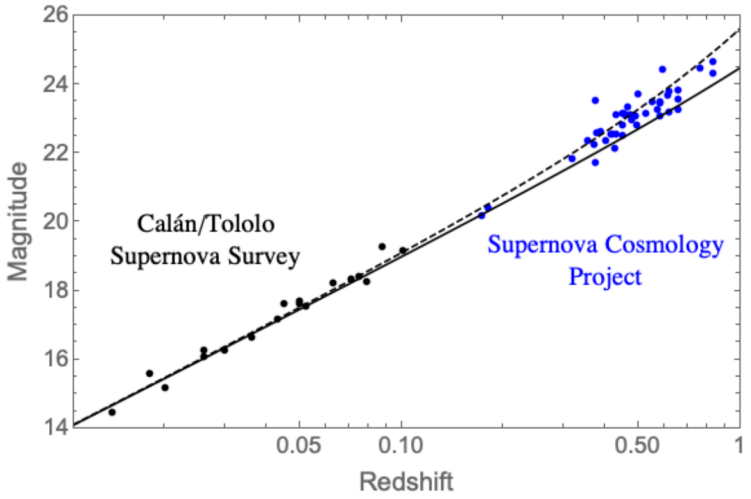


Figure 2: Data from the Supernova Cosmology Project showing deviation from Hubble's law at very large distances.

8. In Figure 2, a higher redshift corresponds to an object that existed earlier in the Universe; therefore, moving from high redshift to low redshift shows a progression from past to present. Given this, how has the Hubble constant (the slope of the best-fit line) evolved over time? Remember that the axes here are *reversed* with respect to our previous plots!

This is a tricky plot to analyze, but you should have concluded that H_0 is getting *larger* over time – that is, the Universe's expansion is *accelerating*. While this discovery earned Perlmutter, Schmidt, and Riess the 2011 Nobel Prize in Physics, we still don't actually know *why* the Universe is accelerating. We therefore call the mysterious force accelerating the Universe **dark energy**.

It turns out that dark energy dominates the energy content of the Universe (68%!!, normal matter is only about 5%). Matter and dark energy are locked in an endless struggle to control the fate of the Universe: while matter wants to pull things together and slow the Universe's expansion, dark energy wants to push the Universe apart.

9. Cosmologists believe that dark energy will forever remain dominant over matter. If this is the case, what will be the ultimate fate of the Universe?

10. If, somehow, matter were to become dominant over dark energy, how would this affect the ultimate fate of the Universe?

3 A Stairway to Heaven: The Cosmic Distance Ladder

In the previous section, you should have found that Hubble's original data had a significant amount of scatter. One major source of error in Hubble's data was that the galaxies he studied were *too close* to us; in the local Universe, gravitational interactions between galaxies compete against the

expansion of space, causing deviations from Hubble's law. Therefore, to ensure that our measured velocities are due solely to the Universe's expansion, we must look at very distant galaxies that are free from the gravitational influence of the Milky Way and its neighbors. We say that these distant galaxies move purely with the "Hubble flow."

Unfortunately, measuring large distances in astronomy is *difficult*. But, with a technique known as the **distance ladder**, we can use measurements of shorter distances to gradually build up to measurements of huge, cosmological distances (which we can then plug into Hubble's law). For instance, we can use *parallax* to find the distances to nearby Cepheid variable stars; we can use nearby Cepheid variables to calibrate the Cepheid *period-luminosity* relationship, giving us the distances to farther Cepheids; we can use the distances of Cepheids in faraway galaxies to calibrate the light curves of distant supernovae; and we can use these supernova light curves to estimate the Hubble constant.

In this section, you will learn about a few key techniques that form the "rungs" of our cosmic distance ladder.

3.1 Parallax

Parallax is the change in the *apparent position* of an object when viewing it from *two different lines of sight* (i.e., perspectives). An illustration of this phenomenon is shown in Fig 3 (left), where the object covers the distant blue square when viewed from "Viewpoint A" but instead covers the distant red square when viewed from "Viewpoint B".

It turns out that this geometric phenomenon is a powerful tool for measuring the distance to relatively nearby stars. Figure 3 (right) shows how parallax works in the context of astronomy. When viewed from point 1, the star at the tip of the triangle will look like it as at position B, while if it is viewed from point 2 it will look like it is at position A. Let's define D to be the distance between the Sun and the other star, B to be the "baseline" (the distance between point 1 and point 2), and θ to be the "parallax angle" (defined as *half* the angle separating A and B of the object). Notice that, in Figure 3 (right), the star, the Sun, and point 1 create a right triangle. Since $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{B/2}{D}$, you can obtain an expression for the distance, D, as

$$D = \frac{B/2}{\tan\theta} \approx \boxed{\frac{B/2}{\theta}}, \tag{2}$$

where we replaced $\tan\theta$ with θ, since these angles are typically quite small (you can check for yourself that $\tan\theta$ is very close to θ for very small θ).

Using Equation 2, **answer the following questions in your lab write-up:**

11. For an object at a fixed distance, how does θ change as you increase B?
12. For a fixed B, how does θ change as D increases?

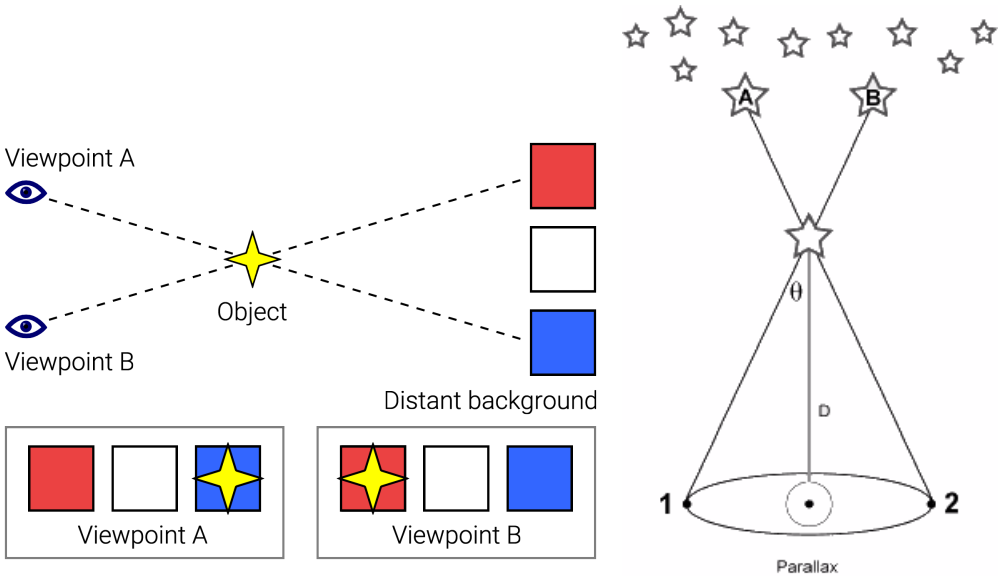


Figure 3: (*Left*) A simplified illustration of the parallax of an object against a distant background due to a perspective shift. (Credit: Wikipedia CC BY-SA 3.0). (*Right*) Parallax in the context of astronomy. Points 1 and 2 indicate positions of the Earth 6 months apart, with the circled dot representing the Sun. Parallax in the context of astronomy. Points 1 and 2 indicate positions of the Earth 6 months apart, with the circled dot representing the Sun.

3.1.1 Parallax in Action: The Gaia Space Observatory

While the parallax technique can be used to infer the distances of relatively nearby stars in the Universe, parallax measurements at astronomical scales are difficult as they yield extremely small angles. Luckily, for an object at a fixed distance, the parallax angle can be increased by using a *longer baseline*.

The European Space Agency's **Gaia space observatory** – a highly successful ongoing mission measuring the parallax of around 2 *billion* stars in our Galaxy (about 2% of the galactic stellar population!) – is able to achieve a baseline equal to the diameter of the Earth's orbit around the Sun (2 AU) by orbiting the Sun alongside the Earth. Gaia is able to precisely measure parallax angles down to 20 μas (micro-arcseconds). An arcsecond (as or $''$) is an astronomical unit of angle, where 1 degree = 60 arcminutes and 1 arcminute = 60 arcseconds. In other words, 1 arcsecond = 1/3600 degrees. Since $\mu = 10^{-6}$, Gaia's precision is 20 μas $\approx 5.55 \times 10^{-9}$ degrees! That's a small angle!!

Answer the following questions in your lab notebook:

13. Suppose you measure a parallax angle of $1''$ using a baseline of $B = 2$ AU (1 AU $\approx 1.496 \times 10^{11}$ m ≈ 8 light−minutes). Using Equation 2, calculate the distance, D, to the object. Did you get something close to $206,265$ AU? This is how we define the **parsec** (or pc) – the distance that produces a *parallax of one (arc)second*.

14. By using parsecs for distance (D) and arcseconds for the parallax angle (θ), we can define a simpler relationship for parallax distance: $D = 1/\theta$. Using this formula and the fact that Gaia can measure parallaxes down to 20 μas, what's the furthest distance Gaia can see? How does this distance compare to the distances you saw in Section 2?

3.2 Cepheids

Unfortunately, the parallax method will not allow us to reach the cosmological distances we need to infer the Hubble constant, H_0; at a certain point, the parallax angles become intractably small (the parallax for the closest star, Proxima Centauri, is already $< 1''$). So, how do we actually obtain distances at large scales? One approach is to use "standard candles," or sources where we know the luminosity (i.e., the intrinsic brightness) *without needing to know the distance*.

Cepheid variable stars are supergiant stars (4-20 times more massive and up to 100,000 times more luminous than our Sun) that periodically change their radius, temperature, and brightness as they radially pulsate. Their potential as standard candles was discovered by the Harvard astronomer Henrietta Swan Leavitt, who in 1908 was able to determine that their luminosity was directly related to their pulsation period (Cepheids with longer periods are intrinsically more luminous). This period-luminosity relationship (the "Leavitt law") means that if one were able to measure the period of the brightness variations of a Cepheid, they could infer the Cepheid's *absolute* brightness (quantified by either the luminosity, L, or the absolute magnitude, M). Since we can readily measure the star's *apparent* brightness (or its apparent magnitude, m), we can use the **distance modulus** to find the distance, d, to the Cepheid. The formula for the distance modulus is given as

$$m - M = 5 \log_{10}(d) - 5, \tag{3}$$

where m is the apparent magnitude, M is the absolute magnitude, and d is the distance (in parsecs).

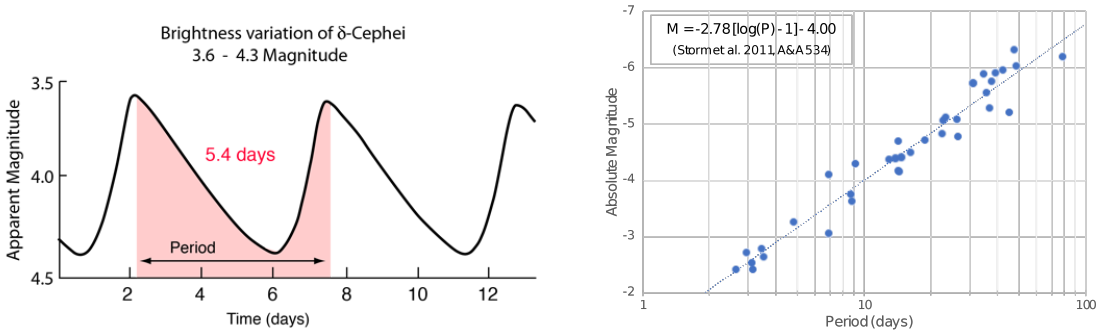


Figure 4: An illustration of the light curve (measurement of brightness over time) of a Cepheid variable *(Left)*. The Period-Luminosity relationship for Cepheid variables *(Right)*.

Answer the following questions in your lab notebook:

15. A measured Cepheid has an apparent magnitude of 12 and a pulsation period of 10 days.

(a) Using the period-luminosity relationship given in Figure 4, determine the absolute magnitude of the Cepheid. (reminder: magnitude measurements don't have units)

(b) Use the distance modulus (Equation 3) to determine the distance to this Cepheid variable (in parsecs).

(c) Recall that a distance in parsecs (measured from the Earth) is related to a parallax angle in arcseconds via $D = 1/\theta$. What is the parallax angle of this Cepheid? Can we measure this parallax using Gaia?

16. There are a number of steps/measurements involved in finding the distance to a Cepheid variable. What are some sources of error that could cause uncertainty in a Cepheid distance measurement? Recall that Edwin Hubble himself used Cepheid data in his 1929 paper.

While Cepheid variables are useful standard candles, there are also difficulties associated with measuring their distances. One major issue is that Cepheids are simply not bright enough to be measured at cosmological distances. A brighter alternative are *supernovae*.

3.3 Supernova light curves

A supernova is the violent explosion of a star. Releasing enormous amounts of energy over the period of a few days, the energy output of a typical supernova can be as much as the total energy output of the Sun *over its 10 billion-year lifetime*. They are relatively rare events with astronomers estimating that only one or two supernova will occur in the Milky Way every century. While we won't go into detail about supernova classification, they can be broadly classified based into two types. Here we will focus on Type Ia ("one-A") supernovae which are commonly referred to as standard candles. Type Ia supernovae, regardless of where they occur in the Universe, are believed to have a fairly consistent peak luminosity making them useful as standard candles.

To get a better understanding of how we can use Type Ia supernovae to obtain cosmological distances, we'll use the "Supernova Light Curve Fitting Explorer" contained in the NAAP Labs app. You should already have NAAP Labs downloaded from the Exoplanet Lab; open the app, click on "14. Cosmic Distance Ladder," and select "Supernova Light Curve Fitting Explorer." You should see a plot with magnitude on the vertical axis (absolute magnitude on the left axis and apparent magnitude on the right) and "Days from Peak" on the horizontal axis.

The second tab of "hubble.csv" (the tab labeled "NAAP SNIa") lists 6 supernovae. For each of these supernovae, follow these instructions to fill out the spreadsheet. **Make sure to include your completed spreadsheet and the appropriate graphs in your lab write-up.**

Step 1: At the top right of the NAAP applet screen, select the appropriate supernova from the drop-down menu labeled "select a supernova."

Step 2: Check the "show horizontal bar" box.

Step 3: If you click and drag the white area of the plot, you can drag the supernova data points to align with the standard light curve for a Type Ia supernova. The data points can be moved both horizontally and vertically. Align the blue data points with the red template curve.

Step 4: Align the horizontal bar with the peak of the curve. The horizontal bar can be moved by dragging either numerical indicator on the left or right sides of the plot.

Step 5: Input the values of the absolute and apparent magnitudes from the horizontal bar into the "Distance Modulus Calculator" box at the bottom of the applet and read off the value for the distance, d, on the bottom right. Record the value of the distance (in Mpc, not pc!) on your spreadsheet.

17. Once you've fully filled out your spreadsheet, make a scatter plot of your supernova data with the recession velocity on the vertical axis and the distance on the horizontal axis. As you did in Section 2, fit a line to the data to find H_0. What value do you get for the Hubble constant?

4 Conclusions

Answer the following questions in your lab write-up:

18. Qualitatively, what does Hubble's law tell us? What are the physical implications of Hubble's law: i.e. how does the Hubble constant (H_0) relate to the expansion of the Universe?

19. One big problem in modern cosmology (and one of the biggest problems in all of physics) is that *we don't actually know* the correct value of H_0 – measuring the expansion rate of the Universe in two different ways gives us two *completely* different numbers. This problem is called the **Hubble tension**.

    (a) Read this short article describing the Hubble tension: `https://astrobites.org/2017/10/13/the-hubble-constants/`

    (b) What are the two primary methods by which astronomers measure H_0, and what values of H_0 do each of these methods predict?

    (c) What might we learn by trying to understanding the source of the Hubble tension?

20. Post at least one question that you still have after finishing this lab onto the Ed Discussion board.

21. If you have any feedback on how today's lab was run, or if you have any suggestions for future lab sessions, please let me know!